

Priyanka Chatterjee

Backend Engineer · Go & Python · Event-Driven Systems · Production Reliability · 4+ Years
Kolkata, West Bengal, India

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SUMMARY

Backend engineer with 4+ years designing reliable, scalable **microservices** in **Go** and **Python**, including **event-driven** systems, production monitoring, and database **optimization**. Demonstrated expertise in **AWS** infrastructure, CI/CD automation with **GitHub Actions**, and diagnosing complex production incidents across **distributed systems**. Led the **AWS SDK JavaScript v2-to-v3** migration across 180+ call sites and 11+ services, reducing bundle size and unblocking runtime upgrades; optimized **PostgreSQL** queries powering compliance audits by 40%, cutting audit time from 3 minutes to 1.5 minutes.

TECHNICAL SKILLS

Languages: Python, Go, SQL, Bash, TypeScript, JavaScript (ES6+), HCL

Backend & APIs: Microservices, Event-Driven Architecture, Distributed Systems, Serverless, REST APIs, gRPC, OAuth 2.0 (Auth0), Gin, Express.js, Node.js

Databases & SQL: PostgreSQL, SQL query tuning, Query optimization, CTEs, jsonb, prepared statements, indexing, database migrations, transaction management

Messaging & Streaming: Apache Kafka, consumer groups, dead-letter queues (DLQs), AWS SNS/SQS, event-driven pipelines, asynchronous processing

Cloud & Serverless: AWS (Lambda, EC2, RDS, S3, CloudTrail, GuardDuty, SecurityHub, X-Ray, CloudWatch), Azure, GCP, serverless architecture, multi-cloud

IaC & DevOps: GitHub Actions, CI/CD pipelines, Docker, Terraform, HCL, Infrastructure as Code, automated testing, release automation, multi-distro environments

Observability: Datadog, OpenTelemetry (OTel), logging, tracing, metrics, incident diagnosis, production monitoring

Testing & Quality: Unit testing, integration testing, automated CI testing, code review, smoke tests

Core CS: Data structures and algorithms, object-oriented design, system design, API design, performance optimization

Security & Compliance: Cloud security, IAM, RBAC, OAuth 2.0, authentication and authorization, compliance frameworks (CIS, NIST)

WORK EXPERIENCE

Turbot HQ, Inc. Remote

August 2022 – April 2026

Software Engineer I

- Designed and developed a **REST API** microservice in **Go** using the **Gin** framework with **OAuth 2.0** authentication (Auth0), containerized with **Docker**, and deployed to **AWS EC2** serving multiple internal gRPC clients; handled authentication, request validation, and service-to-service communication for a multi-tenant SaaS platform.
- Built a **Kafka-based event-driven** pipeline for policy-enforcement workflows with asynchronous consumer processing and a dead-letter queue (**DLQ**) for failure handling and manual investigation; ensured exactly-once delivery semantics and improved reliability through robust error handling.
- Optimized complex **PostgreSQL** queries powering enterprise compliance audits — rewriting **AWS IAM** policy-evaluation queries with jsonb array operations, multi-stage CTEs, and condition-aware filtering — improving query performance 40% (cutting audit-report generation from 3 minutes to 1.5 minutes across 10,000+ resources) and eliminating false-positive findings.
- Led the **AWS SDK** for JavaScript v2 to v3 migration across 180+ call sites and 11+ **AWS** services (SecurityHub, CloudWatch, ControlTower, Shield, X-Ray, **EC2**, EFS, and others) in production Node.js **Lambda** functions, reducing bundle size through modular imports and keeping the platform supported past **AWS SDK v2** end-of-life.
- Engineered core features for Tailpipe, a **Go-based observability** CLI on DuckDB + Parquet, lifting log-processing throughput 30% to 1,000+ logs per second by chunking and batching file reads; authored 11 **AWS** log-source plugin tables (CloudTrail, GuardDuty, SecurityHub, ALB/NLB/CLB/ELB access, VPC flow, WAF) with standardized error handling.
- Designed multi-distro Linux smoke tests (CentOS, Ubuntu ARM64, macOS) for **Go** CLIs in the **GitHub Actions** release pipeline, hardening release safety and catching regressions before deployment; unified release workflows, automated Slack notifications, and **Docker** image tagging for four open-source products.
- Conducted **root cause analysis** on a production **incident**, diagnosing a **Go** service stuck in a Kubernetes CrashLoopBackOff; collaborated with the platform team to investigate pod logs, service status, and restore availability, implementing preventive measures to avoid recurrence.

Kredey (Kredey Innovations Pvt. Ltd.) Remote

2021

Full-Stack Website Developer (Intern)

- Used Elasticsearch to support full-text and semantic search over a Firebase NoSQL database, enabling users to discover content efficiently across a document-sharing platform.

PROJECTS

TODO / Tasks REST API *(April 2026)*

API-first REST backend in TypeScript with Bun, Express.js 5, and SQLite (bun:sqlite). JWT authentication (access, ID, and refresh tokens), role-based access control (RBAC), prepared-statement CRUD service layer with proper REST status codes (201/204/400/404), database migrations, and schema validation with Zod. Demonstrates production-grade API design patterns and secure credential management.

Lila Tic-Tac-Toe — Real-Time Multiplayer Game *(2026)*

Go-based Nakama game server with **PostgreSQL**, deployed on Fly.io. Server-authoritative architecture over WebSocket transport, KV-storage-based matchmaking, mode-aware configuration, optimistic UI with validation-failure recovery, and reconnect-into-active-match reliability. Demonstrates real-time system reliability, database transactions, and production deployment practices.

EDUCATION

University of Engineering & Management (UEM) Kolkata, West Bengal, India

August 2019 – May 2023

Bachelor of Technology in Computer Science and Engineering

CGPA: 9.28

Relevant Coursework: Object-Oriented Programming, DBMS, Data Structures and Algorithms, Shell Scripting in Linux, Low-Level Design, High-Level Design